

Cadre of Employee

Question:

Constructor overloading CO1

An organization is maintaining the data of employee according to cadre of employee with following parameters: name, id, designation, salary, promotion status. Apply the constructor overloading to implement it.

Step wise execution

1. The program starts execution from the main() method.

2. "cadre:Entry level" is displayed.

3. Employee object e1 is created with:

name = sahithi

id = 12454657

designation = interns

salary = 15000

promotion_status = false (default value)

4. The four-parameter Employee constructor is called and initializes e1.

5. e1.employeedetails() is called.

6. The details of e1 are displayed:

name: sahithi

id: 12454657

designation: INTERNS

salary: 15000.0

eligible for promotion: no

7. "cadre:mid-level" is displayed.

8. Employee object e2 is created with:

name = meghana

id = 3434567889

designation = Data scientist

salary = 23454650

promotion_status = true

9. The five-parameter Employee constructor is called
and initializes e2.

10. e2.employeedetails() is called.

11. The details of e2 are displayed:

name: meghana

id: 3434567889

designation: DATA SCIENTIST

salary: 23454650.0

eligible for promotion: yes

12. "cadre:executive level" is displayed.

13. Employee object e3 is created with:

name = sangeetha

id = 32452943

designation = Director

salary = 243654670

promotion_status = false

14. The five-parameter Employee constructor is called
and initializes e3.

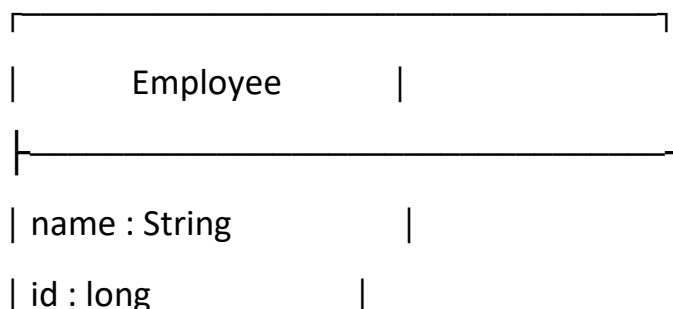
15. e3.employeeDetails() is called.
16. The details of e3 are displayed:
 name: sangeetha
 id: 32452943
 designation: DIRECTOR
 salary: 243654670.0
 eligible for promotion: no
17. After displaying all employee details,
 the program terminates.

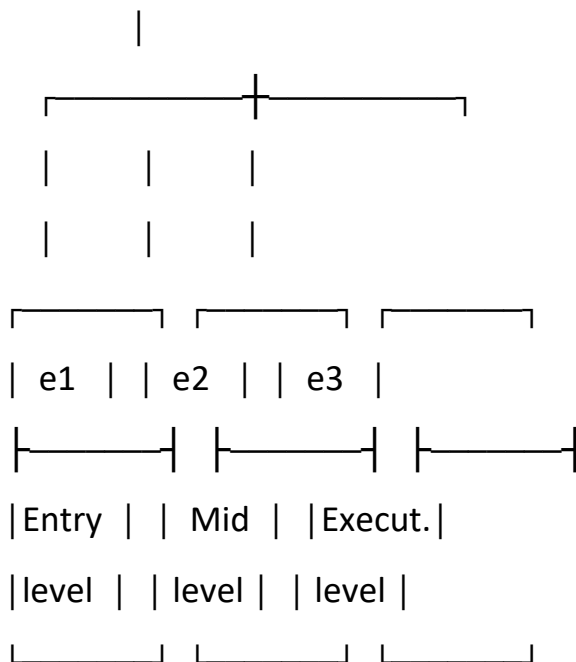
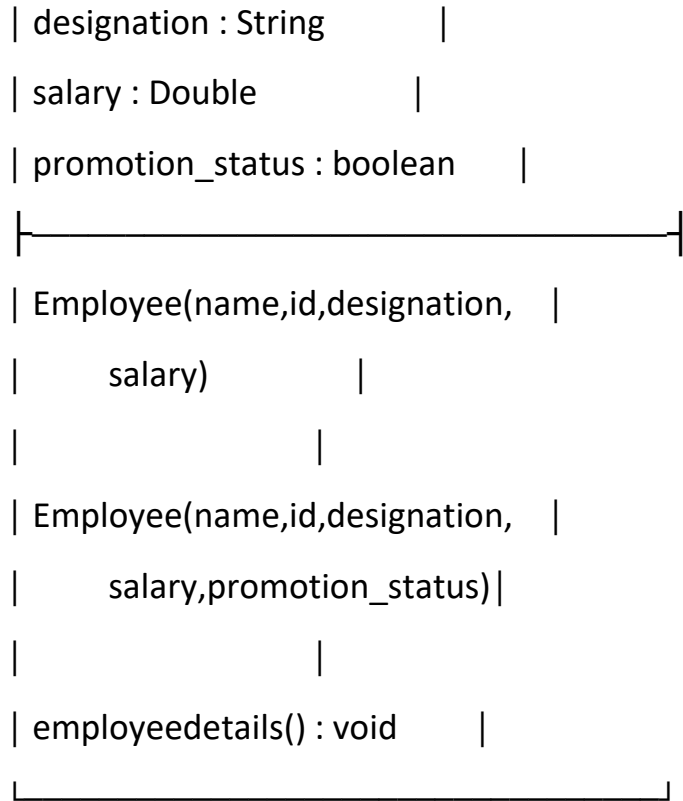
Algorithm

1. Start the program.
2. Define the Employee class with the following data members:
 name, id, designation, salary and promotion_status.
3. Define a constructor with four parameters:
 name, id, designation and salary.
4. Initialize the employee details using the constructor.
5. Define another constructor with five parameters:
 name, id, designation, salary and promotion_status.
6. Initialize all employee details including promotion status.
7. Define the employeeDetails() method to display
 the employee information.
8. Start the main() method.

9. Display "cadre:Entry level".
10. Create employee object e1 using the four-parameter constructor.
11. Call e1.employeedetails() to display the details of the entry-level employee.
12. Display "cadre:mid-level".
13. Create employee object e2 using the five-parameter constructor with promotion status as true
14. Call e2.employeedetails() to display the details of the mid-level employee.
15. Display "cadre:executive level".
16. Create employee object e3 using the five-parameter constructor with promotion status as false.
17. Call e3.employeedetails() to display the details of the executive-level employee.
18. Stop the program.

UML diagram





Code

```
package oops;

class Employee{
    String name;
    long id;
    String designation;
    Double salary;
    boolean promotion_status;

    public Employee(String name,long id,String
designation,Double salary){
        this.name=name;
        this.id= id;
        this.designation= designation;
        this.salary= salary;
    }

    //this is constructor overloading
    public Employee(String name,long id,String
designation,Double salary,boolean promotion_status){
        this.name=name;
        this.id= id;
        this.designation= designation;
        this.salary= salary;
        this.promotion_status=promotion_status;
    }

    void employeeedetails(){
        System.out.println("name:" + name);
        System.out.println("id:" + id);
        System.out.println("designation:" +
designation.toUpperCase());
        System.out.println("salary:" + salary);
    }
}
```

```

        System.out.println("eligible for promotion:" +
(promotion_status ? "yes" : "no") + "\n");
    }
}

public class Cadre_of_Employee {
    public static void main(String[] args) {
        System.out.println("cadre:Entry level");
        Employee e1=new
Employee("sahithi",12454657L,"interns",15000d);
        e1.employeeetails();
        System.out.println("cadre:mid-level");
        Employee e2=new Employee("meghana",3434567889L,"Data
scientist",23454650d,true);
        e2.employeeetails() ;
        System.out.println("cadre:executive level");
        Employee e3=new
Employee("sangeetha",32452943,"Director",243654670d,false);
        e3.employeeetails();
    }
}

```

Output screen shot

```

PS C:\Users\Admin\OneDrive\Documents\basicjava>
PS C:\Users\Admin\OneDrive\Documents\basicjava> c;; cd 'c:\Users\Admin\OneDrive\Documents\basicjava'; & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCo
deDetailsInExceptionMessages' '-cp' 'c:\Users\Admin\OneDrive\Documents\basicjava\bin' 'oops.Cadre_of_Employee'
cadre:Entry level
name:sahithi
id:12454657
designation:INTERNS
salary:15000.0
eligible for promotion:no

cadre:mid-level
name:meghana
id:3434567889
designation:DATA SCIENTIST
salary:2.345465E7
eligible for promotion:yes

cadre:executive level
name:sangeetha
id:32452943
designation:DIRECTOR
salary:2.4365467E8
eligible for promotion:no

```